

KLYUCHI ALTAI, Russian FederationWMO: **360210**Lat: **52.263N**Lon: **79.199E**Elev: **468**StdP: **14.45**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-25.0	-36.2	0.7	-30.8	-29.9	1.0	-25.0	19.2	17.6	17.0	18.3	3.3	270	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	90.3	66.3	87.0	65.3	83.8	64.1	69.7	83.3	68.0	80.5	66.4	78.9	8.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.0	73.9	63.8	90.1	72.4	61.8	83.9	71.5	34.1	82.9	32.7	80.5	31.4	78.7	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.3	14.3	-33.3	96.4	7.0	3.4	-38.3	98.9	-42.4	100.9	-46.3	102.8	-51.4	105.2		
(5)			-34.0	71.9	7.3	3.0	-39.2	74.0	-43.5	75.8	-47.6	77.4	-52.9	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	1.9	6.4	21.7	43.7	56.7	67.2	69.5	65.8	54.4	41.1	22.7	9.5
	DBStd	26.40	15.49	14.04	12.85	10.44	9.30	7.24	5.86	6.90	8.51	9.40	14.32	13.98
	HDD50	6297	1490	1222	877	240	43	1	0	1	49	302	818	1254
	HDD65	10087	1955	1642	1342	641	286	61	26	74	331	742	1268	1719
	CDD50	2125	0	0	0	51	250	518	605	492	183	26	0	0
	CDD65	435	0	0	0	2	27	127	165	99	15	0	0	0
	CDH74	4796	0	0	0	38	446	1380	1523	1165	241	3	0	0
	CDH80	1637	0	0	0	7	129	499	504	422	76	0	0	0
Wind	WSAvg	6.5	5.9	6.1	6.7	7.5	7.5	6.8	6.0	6.0	6.1	6.3	6.5	6.7
Precipitation	PrecAvg	11.0	0.5	0.5	0.6	0.6	0.8	1.4	2.3	1.1	0.8	0.8	0.9	0.7
	PrecMax	13.9	0.9	0.9	1.8	1.5	3.0	4.7	4.5	3.9	2.2	2.5	1.9	1.4
	PrecMin	5.8	0.1	0.0	0.1	0.0	0.2	0.1	0.4	0.0	0.0	0.2	0.2	0.3
	PrecStd	1.9	0.2	0.2	0.4	0.4	0.6	1.0	0.9	0.7	0.5	0.6	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	36.8	55.2	79.2	90.1	93.5	93.1	95.0	88.7	71.9	53.0	37.1
		MCWB	33.8	33.1	45.0	57.6	64.9	66.2	68.9	68.1	63.7	54.3	45.4	34.2
	2%	DB	31.9	33.1	46.8	71.8	83.5	89.2	88.8	89.3	80.6	65.1	47.7	33.6
		MCWB	30.0	30.3	40.4	54.5	61.4	64.3	68.1	65.9	59.7	50.9	41.7	31.2
	5%	DB	27.8	30.2	41.3	66.4	79.5	85.6	85.6	84.6	75.0	60.2	42.9	31.1
		MCWB	25.7	27.6	36.5	51.1	59.1	63.9	66.6	64.2	57.0	48.4	38.2	29.0
	10%	DB	23.7	26.5	37.4	61.3	74.6	82.1	82.3	80.7	69.8	55.0	38.7	28.1
		MCWB	22.0	24.5	34.1	48.5	57.0	63.0	65.3	62.9	54.7	45.8	35.7	26.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	34.2	45.8	59.2	67.0	69.5	73.5	72.0	65.0	55.5	47.1	34.7
		MCDB	35.8	35.9	53.5	73.8	87.2	87.7	85.3	87.0	85.1	68.8	51.2	36.4
	2%	WB	30.1	30.5	40.6	55.3	63.4	67.4	70.6	68.5	60.8	52.1	42.1	31.8
		MCDB	31.8	32.8	46.2	69.6	80.3	82.3	82.6	82.6	76.8	62.8	46.4	33.2
	5%	WB	25.8	27.6	37.0	52.2	60.6	65.9	68.9	66.1	58.3	49.4	38.6	29.1
		MCDB	27.6	29.9	40.8	64.3	75.7	79.8	80.3	80.4	72.2	58.6	42.6	30.7
	10%	WB	22.0	24.5	34.2	49.4	58.3	64.4	67.3	64.3	55.8	46.7	35.8	26.3
		MCDB	23.7	26.5	37.1	60.0	71.7	77.8	78.1	77.8	68.1	54.3	38.7	28.2
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.7	16.5	21.0	22.5	21.2	19.7	21.9	21.0	17.0	13.1	14.8
		MCDBR	16.2	15.9	18.6	28.7	29.9	27.4	25.1	29.1	30.2	25.8	16.0	15.4
		MCWBR	15.2	14.1	12.7	14.8	13.8	10.4	8.8	10.6	13.8	14.2	11.5	14.4
	5% WB	MCDBR	16.7	15.4	17.2	26.2	26.2	22.5	20.4	24.8	26.4	23.1	14.6	15.2
		MCWBR	15.9	14.0	12.4	14.6	13.0	9.8	8.4	9.9	12.5	13.7	11.2	14.8
Clear-Sky Solar Irradiance	taub		0.300	0.336	0.391	0.411	0.415	0.416	0.442	0.420	0.377	0.354	0.339	0.310
	taud		2.028	1.955	1.932	2.146	2.185	2.265	2.200	2.227	2.293	2.256	2.112	2.001
	Ebn at Noon		206	233	245	260	266	266	256	255	252	231	190	174
	Edh at Noon		26	37	47	43	43	41	43	39	33	28	24	22
All-Sky Solar Radiation	RadAvg		397	755	1212	1513	1910	2030	1836	1615	1178	646	329	264
	RadStd		34	36	92	112	135	124	124	101	93	85	34	38